Shipments Analysis

Product Analytics Lead – Aliaa Sakr

Case Study

- Did Bulk Upload reduce churn?
- Does Delivery Window Selection actually work?

Case Context

Two features. Same quarter. Imperfect data.

FEATURE A · WEEK 3

Bulk Shipment Upload

Merchants upload a CSV to create up to 500 shipments at once instead of one by one.

PM's hypothesis:

Saves time for high-volume merchants → reduces churn in the segment most at risk.

FEATURE B · WEEK 8

Delivery Window Selection

After pickup, customers receive an SMS to choose a 4-hour window: morning, afternoon, or evening.

PM's hypothesis:

Customer control reduces failed first-attempt deliveries → lower re-delivery costs.

Dataset: merchants.csv (1,000 rows) shipments.csv (~29,000 rows) events.csv (~38,000 rows)

⚠ *Instrumentation not live on Day 1 of either launch - event data is incomplete for early adopters.*

PART 1

Bulk Upload & Churn

Did it reduce churn? Are the right merchants adopting?

Step 1

Defining Adoption – Merchants buckets

"bulk_upload_completed" event counts are used to segment merchants by engagement depth.

Non-Adopter	Light Adopter	Medium Adopter	Heavy Adopter
597	64	231	108
merchants	merchants	merchants	merchants
59.7%	6.4%	23.1%	10.8%
0 events	*1–2 events*	*3–10 events*	*11+ events*

Key Insight

40% of merchants tried the feature, but only 11% are heavy users (11+ uploads).
Low conversion from trial → habit is a product opportunity - not a failure signal.

Instrumentation gap: Event collection started Jan 20. Early adopters may be under-counted - another reason to prefer depth buckets over binary flags.

Step 2

Churn by Adoption - The Headline Number

Churn rate % is segmented by adoption. Adoption buckets reflects engagement depth.
Usage intensity shows if feature is part of workflow vs. one-time curiosity

Heavy Adopters (11+)	Medium Adopters (3-10)	Light Adopters (1-2)	Non-Adopters
5.6%	**11.3%**	**7.8%**	**28.0%**
churn rate (n=108)	churn rate (n=231)	churn rate (n=64)	churn rate (n=597)

Key Insight:

Churn rate decreases with Bulk Upload Usage intensity.

Statistical significance (Chi- square test):
$\chi^2 = 51.17$
$p < 0.0001$
99.99% confident the pattern is not random.

BUT: correlation ≠ causation. See next slide.

Step 3

Merchant Tiers - The At-Risk Map

Churn rate % is segmented by business tier. Tiers reflect actual pricing, support, and product structure.

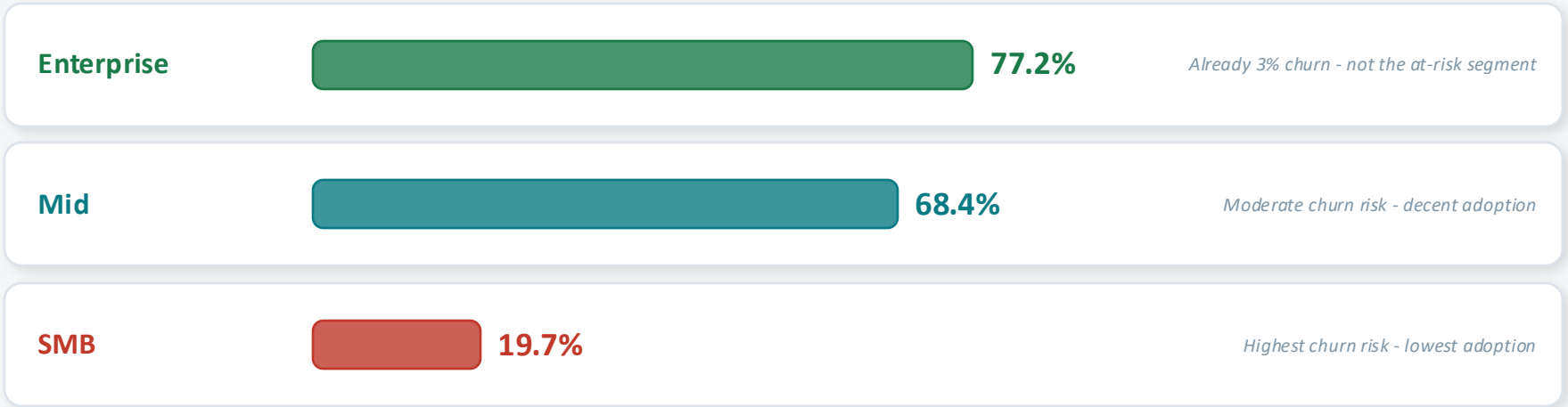
Tier	# Merchants	Avg Volume/mo	Churn Rate	Risk Level
SMB	595 (59.5%)	28.5 shipments	28.7%	HIGH
Mid	304 (30.4%)	273 shipments	9.9%	MODERATE
Enterprise	101 (10.1%)	2,721 shipments	3.0%	LOW

Critical question for the PM hypothesis:

SMB churns at 28.7% - they're the at-risk segment, so, Bulk Upload should help.
If the feature is working, we need to see it reaching and retaining SMB merchants - not just the already-healthy Enterprise tier.

Step 3 & 4

Adoption Skews Toward Already-Healthy Merchants



Adopters' vs Non-Adopters

Metric	Adopters	Non-Adopters	Difference
Avg Monthly Volume	687 shipments	165 shipments	4.2× higher
Enterprise Share	19.4%	3.9%	5× higher
Churn Rate	9.2%	28.0%	3× lower

Step 5

Controlling for Tier & The Error Paradox

Churn by Tier & Adoption Status

If feature CAUSES retention, effect should be uniform across tiers.

Tier	No Adopt	Adopter	Diff
SMB	32.0%	15.4%	-16.6pp
✓ *Real effect where needed most*			
Mid	13.5%	8.2%	-5.4pp
✓ *Real moderate effect*			
Enterprise	4.4%	2.6%	-1.8pp
✗ *Both already healthy*			

⚑ *Effect decreasing SMB→Enterprise suggests growth trajectory, not feature, drives retention.*

The Error Paradox

268 merchants (26.8%) experienced bulk_upload_error events

Merchants with errors: **7.8% churn**
Merchants without errors: **25.0% churn**

Why counterintuitive?

Errors indicate ENGAGEMENT, not failure. Merchants willing to attempt the feature are the same merchants who contact support, iterate, and stick around. Getting errors = trying hard = stickier customer.

This is selection bias again: engagement → adoption + errors + retention.

Conclusion: The effect is real within SMB (-16.6pp) but reverse causality is plausible.
Growing merchants adopt the feature AND stay, not: the feature caused merchants to stay.
Propensity matching or A/B test needed to separate these.

Step 6

Retention Cohorts & Causal Assessment

Retention by Adoption Cohort

Cohort	Retention	N	Avg Usage
Jan 2026 (early)	**92.9%**	98	10.4 events
Feb 2026	**90.7%**	246	7.3 events
Mar 2026 (late)	**87.0%**	54	2.5 events
Apr 2026 (very late)	**100%***	5	1.4 events

**N=5, too small to be meaningful*

Causal Evidence Balance

FOR causality

- ✓ Statistical significance (p<0.0001)
- ✓ Large effect size (18.8pp)
- ✓ Heavy users churn 5.6% (dose-response)
- ✓ Within-SMB effect still -16.6pp
- ✓ High retention across all cohorts (90%+)

AGAINST causality

- Adopters 4.2× higher volume
- Reverse causality (growth → adoption)
- Effect is weakest where merchants are already healthy
- Error paradox (engagement, not feature)

Causal Confidence: 20–30% | Most likely story: Growing merchants adopt the feature as they scale — their retention improvement reflects business growth, not the feature itself.

RECOMMENDATION

Bulk Upload: Where to Invest

Invest for growth & engagement - not as a churn-reduction tool (selection bias too high to claim causality).

LOW PRIORITY

Enterprise

Maintain

Already 77% adoption & 3% churn. No action needed - just keep the feature stable.

MEDIUM PRIORITY

Mid

Optimize

68% adoption, 10% churn. Focus on smoothing the error experience and reducing friction.

HIGH PRIORITY

SMB ★

Investigate & Grow

Only 20% adoption, 29% churn. Survey why they don't adopt (discovery? UX? value?). Target with SMS when volume grows.

Next Steps

Week 1-2: Survey 20 SMB non-adopters (discovery? UX? value?).
Month: **SMS** for SMB when volume grows.
Quarter: Propensity-matched cohort study to isolate true causal effect.

PART 2

Delivery Window Selection

Feature not working - or are we measuring it wrong?

Steps 7, 8, 9, 10 & 12

The Aggregate Is Lying

Pre/post: +3.8% improvement. PM says it barely moved. But the aggregate mixes adopters with non-adopters - we need to separate them.

Pre-Launch (Week 1-7)

70.6%

first-attempt delivery rate
(n=15,798)

—

Post-Launch (Week 8+)

74.4%

first-attempt delivery rate
(n=13,607)

Change

+3.8%

aggregate improvement

Split the post-launch data by adoption:

Customers who CHOSE a window

79.8%

first-attempt rate
(n=6,038)

Customers who did NOT

70.2%

first-attempt rate
(n=7,569)

+9.6%

effect for adopters
$p < 0.0001$

Math check: Effect size (9.6%) × Adoption rate (44.4%) = **4.27% ≈ 3.8% observed.** The numbers reconcile - aggregate dilution explains everything.

Step 13

Effect Consistent Across All Cities

City is the primary segmentation axis - it directly drives delivery logistics (infrastructure, density, courier maturity).
City segmentation is chosen over tier for this analysis.

City	Effect	Adoption	n
Giza	+11.5%	45.7%	1,405
Cairo	+10.2%	43.6%	4,098
Alexandria	+9.9%	45.2%	2,742
Assiut	+9.8%	44.1%	1,328
Suez	+8.8%	45.3%	1,373
Tanta	+8.1%	42.8%	1,348

What this tells us

- Effect range 8-11%: narrow band across very different cities (Cairo dense, Assiut rural)
- Adoption 42-46%: uniform across geographies - bottleneck is not city-specific
- No city shows 0% effect - feature works everywhere, not just well-served areas
- The adoption bottleneck is systematic (likely SMS delivery/discovery), not geographic.

Steps 11 & 14

Who Benefits Most & Controlling for Selection Bias

Feature Effect by Merchant Tier

Secondary lens: Who benefits most from the feature?

Tier	Without	With	Effect
SMB	66.0%	77.4%	+11.5%
Most to gain — weakest baseline			
Mid	74.3%	82.2%	+7.8%
Moderate baseline, solid gain			
Enterprise	82.6%	85.7%	+3.0%
Already optimized — least uplift			

Propensity Score Matching

Counterfactual: Match adopters with non-adopters on same tier + city

Matched adopters: **79.8%**
Matched non-adopters: **80.5%**

Key interpretation:

When matched on tier + city, adopters and non-adopters show nearly identical rates (-0.7%). This means customers who choose delivery windows are not systematically "better" on these observable dimensions - the +9.6% effect is not explained by selection on tier/city alone.

Causal Confidence: 40-50%
7 factors support causality vs 4 against.
Propensity matching holds - selection bias on tier + city is low.
Unmeasured factors (address quality, customer reliability) could still bias.
A/B test would be definitive.

RECOMMENDATION

Delivery Window: Fix Adoption, Not the Feature

Verdict: Feature WORKS (+9.6%, p<0.0001, consistent across all cities).
Problem is adoption (44%), not feature quality.

Week 1-2

Diagnose the bottleneck

- Pull SMS delivery logs (sent? delivered? clicked?)
- Survey 100 non-adopting customers
- "Did you receive SMS?" → "Did you understand it?" → "Why not use?"

Month

Run SMS A/B test

- Current: generic delivery window offer
- Test: "Deliver today 2-5 PM? Choose your slot"
- Goal: click-through → adoption 44% → 50%+

Quarter

Measure ROI

- Adoption goal: 60%+ (60% × 9.6% = 5.8% aggregate)
- Track re-delivery cost savings alongside rate

Impact formula: Real Impact = Effect Size × Adoption Rate. Current: 9.6% × 44% = 4.2% | Target: 9.6% × 60% = 5.8% | Stretch: 9.6% × 80% = 7.7%

Summary

Two Features. Clear Verdicts.

BULK UPLOAD

Adopters churn 9.2% vs 28% non-adopters
(18.8pp difference, p<0.0001)

Finding

Selection bias likely dominates. Adopters are 4.2× higher volume - they were healthier to start with. Effect within SMB is real (-16.6pp) but not purely causal.

Confidence

20–30% causal confidence

Recommendation

Invest for growth.
Fix SMB discovery (survey).
Don't claim churn causality.

DELIVERY WINDOW

Adopters: 79.8% first-attempt vs 70.2%
(+9.6pp, p<0.0001, consistent all cities)

Finding

Feature works. Aggregate looks flat (+3.8%) because only 44% of customers use it. Effect is uniform across all cities - adoption bottleneck is systematic (SMS/discovery).

Confidence

40–50% causal confidence

Recommendation

Keep feature.
Fix adoption via SMS A/B test.
Target 60% → 5.8% aggregate impact.